



UM HACKATHON 2026



URBAN VERTICAL FARM AUTOMATION SYSTEM

Exploring innovations that will shape our world tomorrow.

Start



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PROBLEM STATEMENT



Limited Urban Space

Urban areas lack sufficient land for traditional farming

Resource Waste

Traditional irrigation wastes up to 30% of water & fertilizer

Inefficient Monitoring

Manual farming is time-consuming and prone to human error

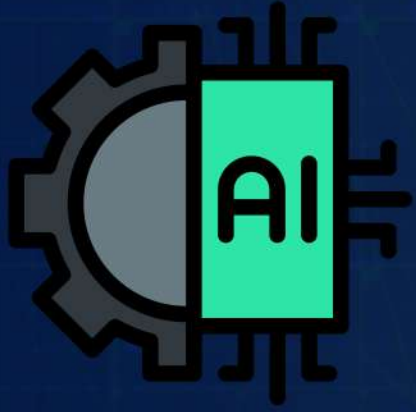
Rising Food Demand

Growing population increases pressure on food production





ArgiMind



OUR SOLUTION: AGRIMIND

AI-assisted precision agriculture for urban vertical farms

An automated system that delivers the right resources, at the right time, with zero waste.

- • Live farm sensing
- • AI recommendation powered by Gemini
- • OpenWeather-supported decision context
- • Dashboard-based monitoring and action explanation

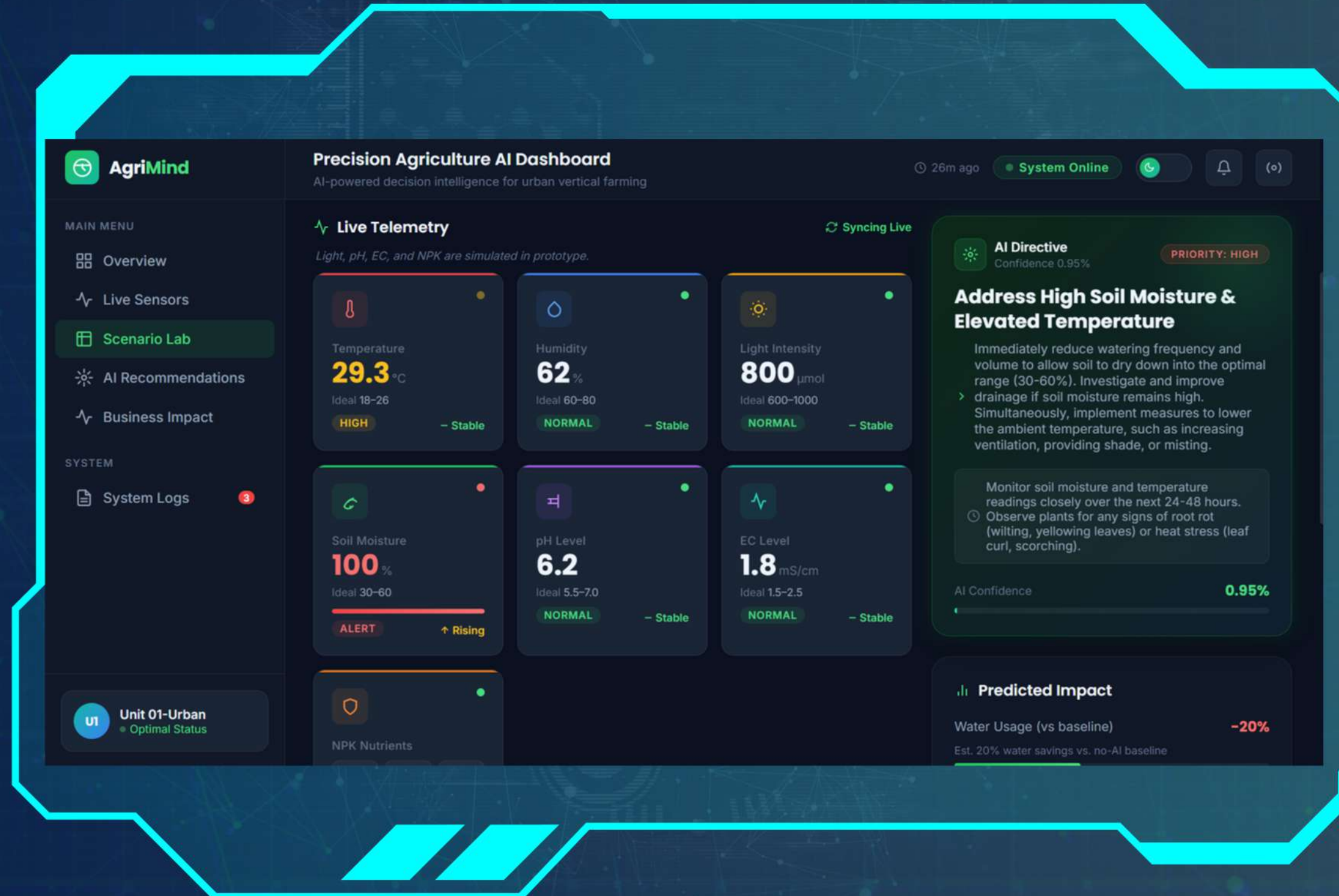
Right data, right action, less waste.





ArgiMind

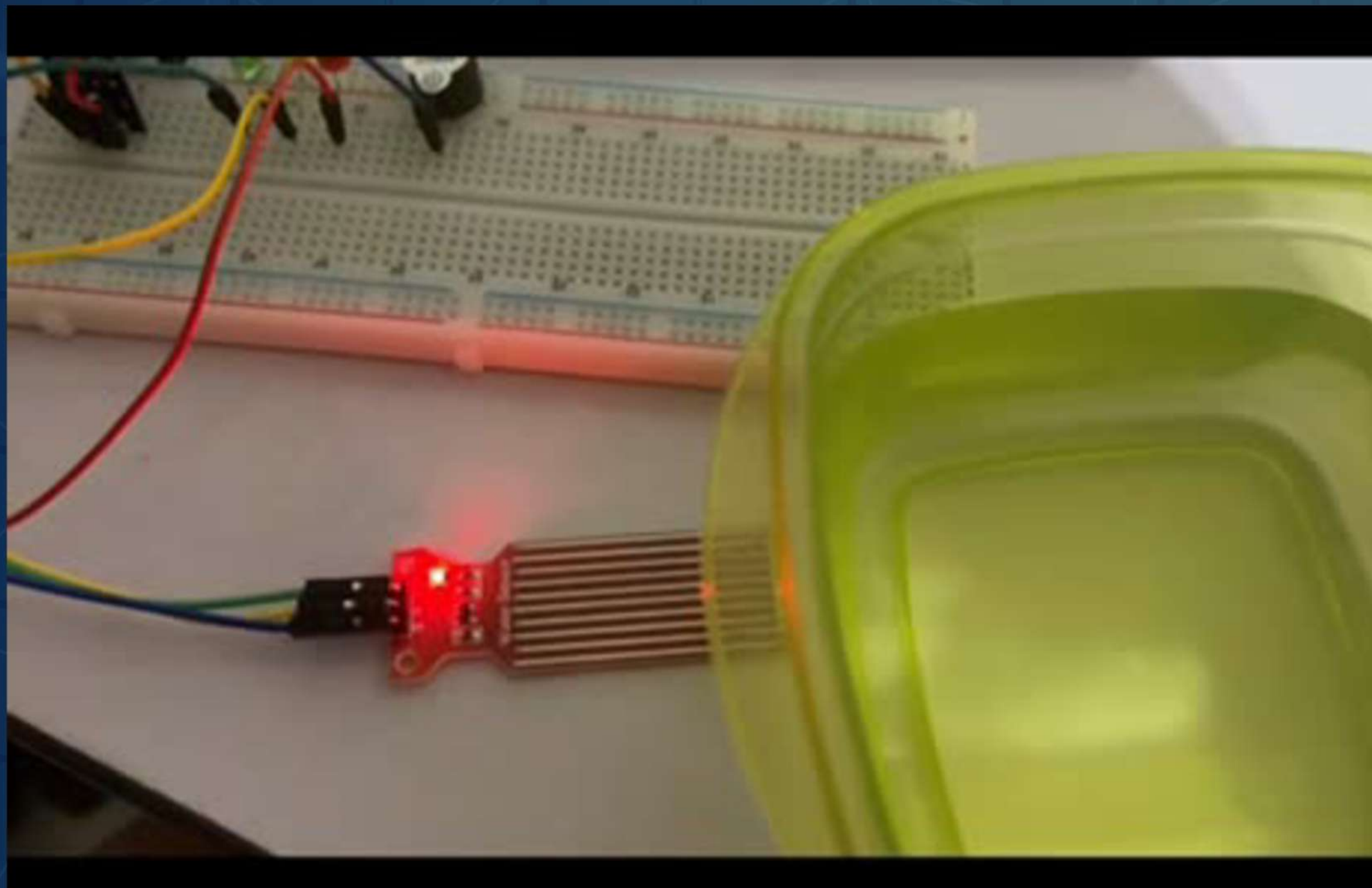
SYSTEM OVERVIEW





ArgiMind

PROTOTYPE DEMONSTRATION





CORE SYSTEM FEATURES



1.

Live Sensor
Monitoring

3.

Gemini AI
Recommen
dation
Engine

5.

Dashboard
& System
Logs

2.

Smart
Irrigation
Decision
Support

4.

OpenWeather
Context
Integration



PROJECTED IMPACT & VALUE

1.

Reduces unnecessary watering

AI recommends short-cycle irrigation instead of fixed timed watering.

3.

Supports resource optimization

Water, lighting, and nutrients can be adjusted based on sensor conditions.

4.

Reduces crop risk

Early alerts help prevent heat stress, dry soil, and poor growing conditions.

5.

Supports urban food security

Helps vertical farms operate more efficiently in limited urban spaces.

2.

Improves decision speed

Farm operators get instant recommendations with clear reasoning.





Target Customers

- Urban vertical farms
- Greenhouse operators
- Smart agriculture companies

Business Model

- SaaS-based precision agriculture system
- Subscription service (monthly / yearly)
- Data-driven farming optimization

Revenue Streams

- Subscription fees
- Hardware setup (initial installation)
- Maintenance & support services



From traditional farming → to Agriculture-as-a-Service



FUTURE SCALABILITY

- ESP32 / MQTT wireless sensor deployment
- Real NPK, pH, EC, and light sensor integration
- Database storage for historical sensor and AI decision logs
- Mobile app for remote monitoring and alerts
- AI crop prediction based on historical trends
- Automated pump, lighting, and nutrient control





THANK YOU!

As we look to the future, these technologies will play a crucial role in shaping a better world for generations to come.

